

Customer & Marketing Analytics

Group Work #1

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1. Predicting Engagement Rate

For Question #1, we decided to use a Linear Regression model. The predictors we chose were the columns:

- **'Topic_Promo_Deal'** & **'Topic_CTA'** to see if "hard-sell" tactics actually drive community engagement or if they potentially turn off the audience
- **'Platform'** & **'Post_Type'** to identify which technical environments (TikTok vs. IG) and formats (Video vs. Static) are currently trending with users
- **'Influencer_Tier'** to test the theory that smaller, niche creators foster higher engagement than broad-reach celebrities
- **'Caption_Sentiment'** measures the emotional impact of the brand's voice on audience reaction
- **'Product_Category'** which has been broken down into two dummy variables:
 - **'ProdCat_Fashion'**
 - **'ProdCat_Tech'**
 - used as control variables to account for the baseline differences in how audiences interact with different product types (e.g., Tech vs. Beauty)

1. Predicting Engagement Rate

When we ran the OLS regression, the statistics that stood out the most were:

- **‘Platform’** coef: 2.4565, meaning that moving from Instagram to Tiktok adds 2.45% to the engagement rate. Tiktok is a massive multiplier for engagement in this campaign
- **‘Post_Type’** coef: 1.9075, meaning that moving from static photos to short-form videos adds 1.91% to the engagement rate. “Video First” strategy is non-negotiable for Pulse Digital
- **‘Influencer_Tier’** coef: -1.7455, meaning that macro-influencers have an engagement rate 1.75% lower than micro-influencers which suggests that for Pulse Digital, smaller creators are more effective at driving community interaction

Big Takeaway: ‘Promo Deals’ and ‘CTAs’ are not that great for likes.

ALSO R-Squared of 0.592 indicates that our chosen variables explain ~59% of the variance in Engagement Rate, representing a high-quality model for social media data

2. Predicting Click-Through

For Question #2, we decided to use a Logistic Regression model. The predictors we chose were the columns:

- **‘Topic_Promo_Deal’, ‘Topic_CTA’** as content pillars
- **‘Platform,’ ‘Post_Type’** as technical format
- **‘Influencer_Tier’** as creator strategy
- **‘Caption_Sentiment’** as audience strategy
- **‘ProdCat_Fashion’ & ‘ProdCat_Tech’** as controls

The objective being: To determine which specific content features move a user from scrolling to clicking

2. Predicting Click-Through

- ‘Topic_CTA’ (Odds Ratio: 3.09): having a Call to Action (like "Shop Now") makes a user **3 times (309%) more likely to click** than a post without one
- ‘Topic_Promo_Deal’ (Odds Ratio: 1.61): mentioning a deal **increases the odds of a click by 61%**
- ‘Post_Type’ (Odds Ratio: 1.52): videos are still better than photos, **increasing click odds by 52%**
- ‘Platform’ (Odds Ratio: 1.19): TikTok is still better for clicks than Instagram, but only by about 19%

Big Takeaway: Pulse should use positive, lifestyle-focused video content with Micro-influencers to build brand engagement, but use Macro-influencers with clear ‘Promo Deals’ and ‘CTAs’ when the goal is immediate website traffic.

3. Predicting Comment Count

For Question #3, we decided to use a Poisson Regression model. The predictors we chose were the columns:

- **'Topic_Promo_Deal' & 'Topic_CTA'** to analyze if promotional content or direct calls to action drive higher conversation volumes
- **'Platform' & 'Post_Type'** to identify if TikTok vs. Instagram or Video vs. Static formats are more effective at generating comments
- **'Influencer_Tier'** to determine if the size of the creator's following impacts the total number of comments received
- **'Caption_Sentiment'** to see if the emotional tone of the caption encourages users to respond
- **'ProdCat_Fashion' & 'ProdCat_Tech'** control variables to account for baseline differences in engagement across different industries

3. Predicting Comment Count

- Comparing Poisson and Negative Binomial (NBD) models, we selected the Negative Binomial model (Log-Likelihood: -2816.0) as it better accounts for the high variance and "viral" spikes in comment counts
- **Biggest Driver of Conversation:** Influencer_Tier (Coef: 0.7446)
 - Macro-influencers drive significantly more total comments than micro-influencers
- Platform (Coef: 0.7707)
 - shows that TikTok is a massive driver for comment volume compared to Instagram
- Topic_CTA (Coef: 0.3219) is a highly significant driver for comments
 - asking for engagement actually works to increase conversation volume.
- Topic_Promo_Deal (p-values 0.873) and Caption_Sentiment (p-value 0.117)
 - while sentiment helps "likes" and deals help "clicks", they do not significantly increase the number of comments left on a post

Big Takeaway: Pulse Digital should utilize **Macro-influencers** on **TikTok** with **specific Calls to Action (CTAs)** if the campaign goal is to maximize community conversation and comment volume.

4. Interaction Effects

For Question #4, we decided to use a Multivariate Logit model. The predictors we chose were the columns:

- 'Topic_Promo_Deal', 'Topic_CTA', 'Platform', 'Post_Type', 'Influencer_Tier', 'Caption_Sentiment', 'ProdCat_Fashion', 'ProdCat_Tech' as main effects
- 'CTA_x_Platform' the interaction term created by multiplying the 'CTA' variable by the 'Platform' variable

We tested whether the effectiveness of a CTA was moderated by the Platform. We hypothesized that TikTok's fast-paced environment might require a stronger CTA "boost" to drive clicks compared to Instagram.

4. Interaction Effects

- **‘CTA_x_Platform’** (p-value 0.299): the interaction effect is statistically insignificant, meaning the power of a CTA to drive clicks is consistent across both platforms
 - we don't need a different CTA strategy for TikTok vs. Instagram
 - a good CTA works effectively on both.
- **‘Topic_CTA’** (Odds Ratio: 3.72): confirms that a CTA is the single most vital component for conversion, regardless of where it is posted

The Pseudo R-Squared remained stable (approx. 0.076), confirming that while the interaction didn't add much explanatory power, our primary drivers (CTA, Promo Deals, and Post Type) remain the most reliable predictors of traffic.

Pulse Digital **BIG TAKEAWAY!!!!**

- Focus on Video content on TikTok using Micro-influencers
 - keep the sentiment positive!!!!
- Macro-influencers are slightly more effective
 - always include a Call to Action and mention Promo Deals
- Use TikTok and Macro-influencers
 - unlike likes, comments are driven by higher-tier creators who can spark a larger discussion

Our interaction analysis proves that a strong Call to Action is a universal win! It works equally well on both TikTok and Instagram.